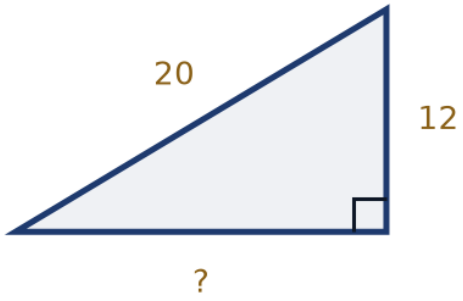
Trigonometry & Additional Topics

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

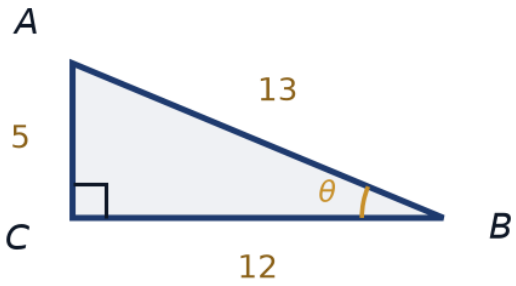
1 [Foundational] • Target: Right-triangle trigonometry

In the right triangle shown below, the hypotenuse is 20 and the side opposite angle θ is 12. Find the length of the adjacent side, then find $\sin\theta$, $\cos\theta$, and $\tan\theta$.



2 [Foundational] • Target: Right-triangle trigonometry

In a right triangle, angle θ has adjacent side 7 and hypotenuse 25. Find $\sin\theta$ and $\tan\theta$.



3 [Foundational] • Target: Right-triangle trigonometry

A 15-foot ladder leans against a wall, making a 50° angle with the ground. Find how high up the wall the ladder reaches, to the nearest tenth of a foot.

4 [Foundational] • Target: The Pythagorean identity

Given $\sin\theta = \frac{5}{13}$ with θ acute, find $\cos\theta$ using $\sin^2\theta + \cos^2\theta = 1$.

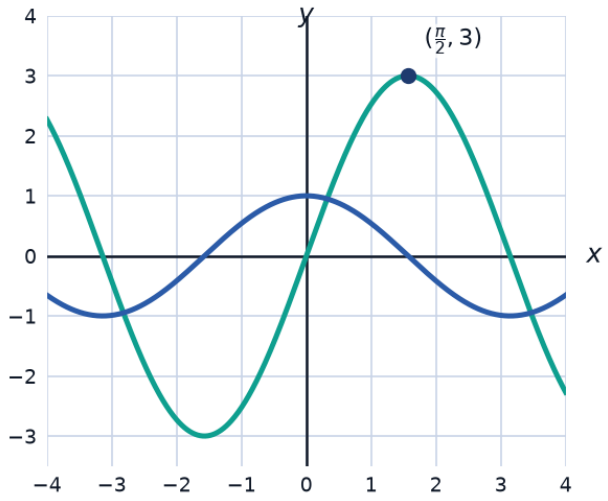
5 [Foundational] • Target: The Pythagorean identity

Given $\cos\theta = \frac{8}{17}$ with θ acute, find $\sin\theta$.

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6 [Foundational] • Target: Exact trigonometric values

State the exact value of $\sin\left(\frac{\pi}{6}\right)$ and $\cos\left(\frac{\pi}{4}\right)$.



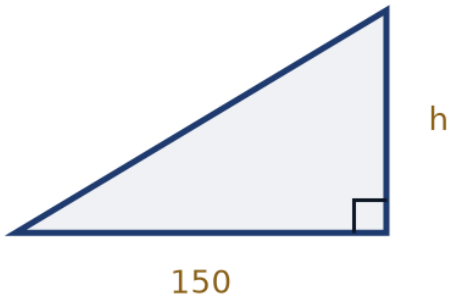
7 [Foundational] • Target: Exact trigonometric values

State the exact value of $\tan(60^\circ)$ and $\sin(90^\circ)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Solving for missing sides and angles

A surveyor stands 150 feet from the base of a tower and measures the angle of elevation to the top as 32°, as shown below. Find the height of the tower, to the nearest foot.



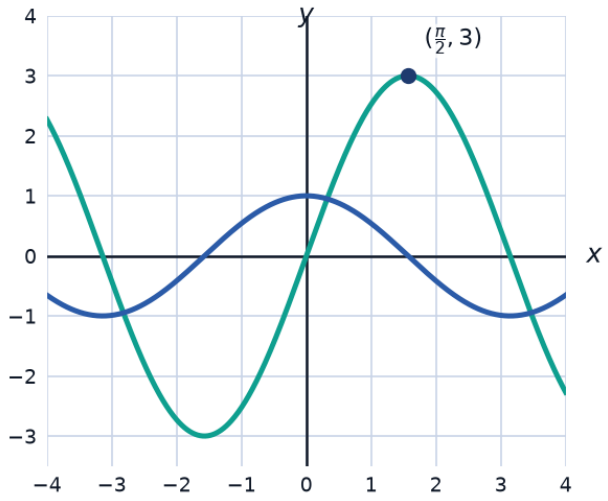
9 [Intermediate] • Target: Solving for missing sides and angles

From the top of a 60-foot cliff, the angle of depression to a boat is 25°. Find the horizontal distance from the base of the cliff to the boat, to the nearest foot.

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10 **[Intermediate]** • **Target: Complex numbers: arithmetic**

Add $(4 + 3i) + (2 - 7i)$.



11 **[Intermediate]** • **Target: Complex numbers: arithmetic**

Multiply $(3 + 2i)(1 - 4i)$.

12 **[Intermediate]** • **Target: Complex numbers: arithmetic**

Simplify i^7, using $i^2 = -1$.

13 **[Intermediate]** • **Target: Complex numbers: modulus and conjugates**

Find the modulus $|6 - 8i|$.

14 **[Intermediate]** • **Target: Complex numbers: modulus and conjugates**

Find the complex conjugate of $-3 + 5i$, and compute the product of the number and its conjugate.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

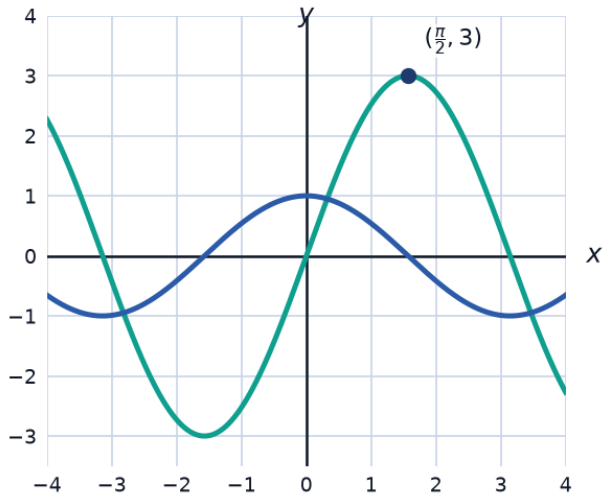
15 **[Challenging]** • **Target: Vectors: magnitude and operations** • **Cross-Lesson: 8.01 Exponentials & Logarithms**

Vector $v = (3, 4)$. Find its magnitude.

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16 [Challenging] • Target: Vectors: magnitude and operations • Cross-Lesson: 1.01 Heart of Algebra

Given vectors $u = (2, -5)$ and $v = (-6, 3)$, find $u + v$.



17 [Challenging] • Target: Vectors: magnitude and operations • Cross-Lesson: 2.01 Quadratics & Polynomials

Given vector $w = (8, -6)$, find $2w$ and the magnitude of $2w$.

18 [Challenging] • Target: Radian measure • Cross-Lesson: 3.01 Functions & Graphs

Convert 150° to radians, in terms of π.

19 [Challenging] • Target: Radian measure • Cross-Lesson: 4.01 Ratios, Rates & Data

Convert $\frac{7\pi}{4}$ radians to degrees.

20 [Challenging] • Target: A well-designed word problem: measuring the width of a river • Cross-Lesson: 5.01 Statistics & Probability

This question has three parts. A surveyor stands at point A directly across a river from a tree at point B (so angle $A = 90^\circ$), then walks 80 meters along the riverbank to point C. (a) At point C, the angle between the bank and the line of sight to the tree is 35°. Use the tangent ratio to find AB, the width of the river, to the nearest meter. (b) Find the direct distance CB from C to the tree, to the nearest meter. (c) A colleague measures from a point 50 meters past C (so 130 meters from A) and finds the angle to the tree is 24°. Use this second measurement to find a second estimate of AB, to the nearest meter, and state which measurement gives the larger estimate.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Right-triangle trigonometry

In a right triangle, the hypotenuse is 87 and the side opposite angle θ is 60. Find $\sin\theta$.

a. $\frac{21}{29}$

b. $\frac{20}{29}$

c. $\frac{20}{21}$

d. $\frac{62}{87}$

22 [Foundational] • Target: The Pythagorean identity

In a right triangle, the hypotenuse is 50 and the side opposite angle θ is 14. Find $\sin\theta$.

a. $\frac{8}{25}$

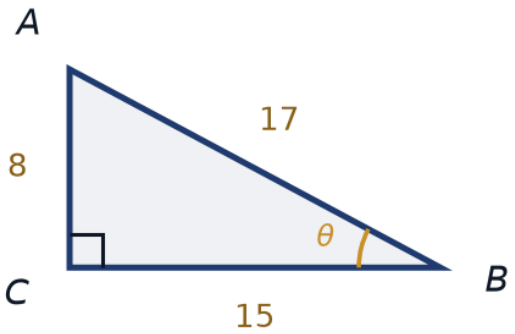
b. $\frac{7}{25}$

c. $\frac{24}{25}$

d. $\frac{7}{24}$

23 [Foundational] • Target: Exact trigonometric values

In a right triangle, the hypotenuse is 25 and the side adjacent to angle θ is 24. Find $\cos\theta$.



a. $\frac{26}{25}$

b. $\frac{24}{25}$

c. $\frac{7}{24}$

d. $\frac{7}{25}$

24 [Foundational] • Target: Solving for missing sides and angles

In a right triangle, the side opposite angle θ is 40 and the side adjacent is 42. Find $\tan\theta$.

a. 1

b. $\frac{21}{29}$

c. $\frac{20}{29}$

d. $\frac{20}{21}$

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25 [Foundational] • Target: Complex numbers: arithmetic

In a right triangle, the hypotenuse is 15 and the side opposite angle θ is 9. Find $\sin\theta$.

a. $\frac{3}{4}$

b. $\frac{11}{15}$

c. $\frac{4}{5}$

d. $\frac{3}{5}$

26 [Foundational] • Target: Complex numbers: modulus and conjugates

In a right triangle, the hypotenuse is 51 and the side opposite angle θ is 24. Find $\sin\theta$.

a. $\frac{15}{17}$

b. $\frac{26}{51}$

c. $\frac{8}{15}$

d. $\frac{8}{17}$

27 [Foundational] • Target: Vectors: magnitude and operations

In a right triangle, the hypotenuse is 10 and the side adjacent to angle θ is 8. Find $\cos\theta$.

a. 1

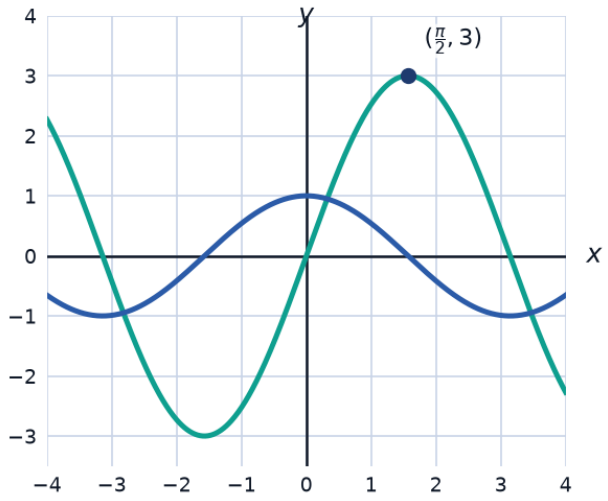
b. $\frac{4}{5}$

c. $\frac{3}{4}$

d. $\frac{3}{5}$

28 [Foundational] • Target: Radian measure

Given $\sin\theta = \frac{11}{61}$ with θ acute, find $\cos\theta$ using $\sin^2\theta + \cos^2\theta = 1$.



a. $\frac{62}{61}$

b. $\frac{60}{61}$

c. 1

d. $\frac{11}{61}$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: A well-designed word problem: measuring the width of a river**

Given $\sin\theta = \frac{9}{41}$ with θ acute, find $\cos\theta$ using $\sin^2\theta + \cos^2\theta = 1$.

a. $\frac{40}{41}$

b. $\frac{9}{41}$

c. 1

d. $\frac{42}{41}$

30 **[Intermediate]** • **Target: Trigonometric Ratios & Cofunctions**

Given $\cos\theta = \frac{60}{61}$ with θ acute, find $\sin\theta$ using $\sin^2\theta + \cos^2\theta = 1$.

a. $\frac{11}{60}$

b. $\frac{60}{61}$

c. $\frac{12}{61}$

d. $\frac{11}{61}$

31 **[Intermediate]** • **Target: Unit Circle & Radians**

Given $\cos\theta = \frac{35}{37}$ with θ acute, find $\sin\theta$ using $\sin^2\theta + \cos^2\theta = 1$.

a. $\frac{12}{37}$

b. $\frac{35}{37}$

c. $\frac{13}{37}$

d. $\frac{1}{3}$

32 **[Intermediate]** • **Target: Complex Numbers & Arithmetic**

Given $\sin\theta = \frac{20}{29}$ with θ acute, find $\cos\theta$ using $\sin^2\theta + \cos^2\theta = 1$.

a. $\frac{22}{29}$

b. $\frac{20}{29}$

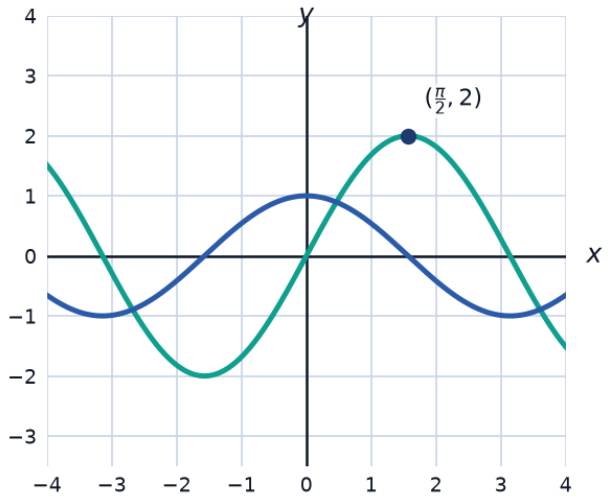
c. $\frac{21}{29}$

d. $\frac{3}{4}$

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33 [Intermediate] • Target: Coordinate Geometry of Circles

Given $\sin\theta = \frac{3}{5}$ with θ acute, find $\cos\theta$ using $\sin^2\theta + \cos^2\theta = 1$.



a. $\frac{4}{5}$

b. $\frac{3}{5}$

c. $\frac{6}{5}$

d. 1

34 [Intermediate] • Target: Right-triangle trigonometry

Multiply $(8 - i)(1 + 5i)$.

a. $16 + 39i$

b. $13 + 39i$

c. $3 + 39i$

d. $13 + 41i$

35 [Intermediate] • Target: The Pythagorean identity

Multiply $(-1 - 9i)(5 + 2i)$.

a. $13 - 47i$

b. $-23 - 47i$

c. $13 + 43i$

d. $13 - 45i$

36 [Intermediate] • Target: Exact trigonometric values

Multiply $(7 + 3i)(-1 - 8i)$.

a. $-31 - 59i$

b. $17 - 59i$

c. $17 - 53i$

d. $17 - 57i$

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37 [Intermediate] • Target: Solving for missing sides and angles

Given vectors $u = (9, -8)$ and $v = (5, -3)$, find $u + v$.

a. $(-11, 14)$

b. $(14, -11)$

c. $(14, -13)$

d. $(16, -11)$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Complex numbers: arithmetic • Cross-Lesson: 3.01 Functions & Graphs

Given vectors $u = (-4, -2)$ and $v = (0, 5)$, find $u + v$.

a. $(-4, 1)$

b. $(3, -4)$

c. $(-2, 3)$

d. $(-4, 3)$

39 [Challenging] • Target: Complex numbers: modulus and conjugates • Cross-Lesson: 4.01 Ratios, Rates & Data

Convert 120° to radians, in terms of π.

a. $\frac{4\pi}{3}$

b. $\frac{\pi}{2}$

c. $\frac{2\pi}{3}$

d. π

40 [Challenging] • Target: Vectors: magnitude and operations • Cross-Lesson: 5.01 Statistics & Probability

Convert 345° to radians, in terms of π.

a. $\frac{23\pi}{13}$

b. $\frac{23\pi}{6}$

c. $\frac{23\pi}{12}$

d. 2π

41 [Challenging] • Target: Trigonometric Ratios & Cofunctions • Cross-Lesson: 6.01 Geometry

If $\sin(x^\circ) = \cos(42^\circ)$, where $0 < x < 90$, what is the value of x?

a. 42

b. 48

c. 52

d. 138

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42 [Challenging] • Target: Unit Circle & Radians • Cross-Lesson: 8.01 Exponentials & Logarithms

Convert 210° to radians.

a. $\frac{5\pi}{6}$

b. $\frac{7\pi}{6}$

c. $\frac{4\pi}{3}$

d. $\frac{11\pi}{6}$

43 [Challenging] • Target: Unit Circle & Radians • Cross-Lesson: 1.01 Heart of Algebra

If θ is in Quadrant IV and $\cos\theta = \frac{3}{5}$, what is $\sin\theta$?

a. $-\frac{4}{5}$

b. $\frac{4}{5}$

c. $-\frac{3}{4}$

d. $\frac{3}{4}$

44 [Challenging] • Target: Complex Numbers & Arithmetic • Cross-Lesson: 2.01 Quadratics & Polynomials

What is the simplified form of $\frac{10}{3-i}$?

a. $3+i$

b. $3-i$

c. $\frac{3+i}{2}$

d. $3+2i$

45 [Challenging] • Target: Coordinate Geometry of Circles • Cross-Lesson: 3.01 Functions & Graphs

What is the radius of the circle defined by $(x+4)^2 + (y-7)^2 = 81$?

a. 7

b. 9

c. 18

d. 81

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